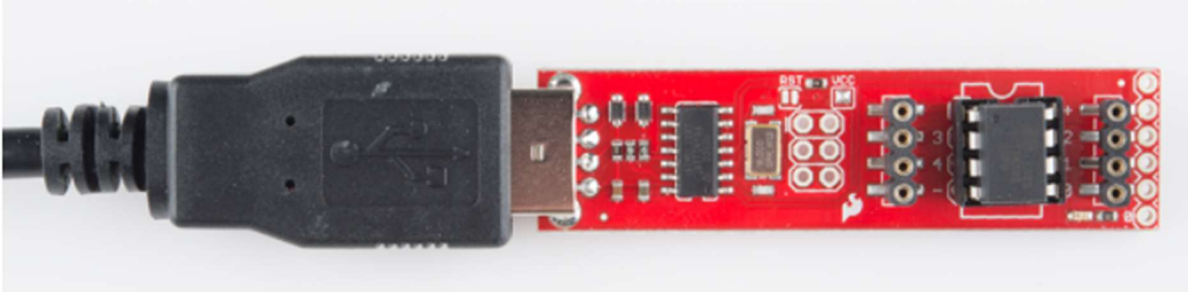


Notes on using the Arduino IDE to program the ATtiny85

July 20, 2026

USING THIS PROGRAMMER:



Unless this is a new Arduino install on the computer using to program, don't mess with the pre-installed drivers. That is not the issue if it fails.

BUT, IF the drivers need to be updated, this works:

To use a SparkFun programmer (like the Pocket AVR or Tiny AVR) on Windows, download and install the **Zadig** driver installer. Plug in your programmer, open Zadig, select **USBtinyISP** from the dropdown, and install the **libusb-win32** driver to establish the connection. [1, 2, 3]

Because SparkFun programmers generally fall into two categories (AVR Programmers or Boards with USB-to-Serial chips), the exact setup process varies:

1. AVR Programmers (Pocket AVR / Tiny AVR)

Used for burning bootloaders or programming raw chips via ISP. [1, 2]

- **Driver:** Windows will likely struggle to recognize this natively. Download and run [Zadig](#), select *libusb-win32* as the driver, and click *Install Driver*. [1, 2, 3, 4]
- **Arduino IDE:**
 1. Go to **Tools > Programmer** and select **USBtinyISP**.
 2. To upload code without a bootloader, click **Sketch > Upload Using Programmer**. [1, 2]
- **Command Line:** Uses `avrdude` natively. [1, 2]

With the correct drivers, the programmer (**FabISP**) will show up like this in the system devices list in the **DEVICE MANAGER**:



USING THE Arduino IDE to program:

SELECT THE FOLLOWING UNDER THE **TOOLS** TAB IN THE IDE:

1. BOARD. Select "ATTiny25/45/85"
2. PORT. Leave this alone!
3. CLOCK: Select "Internal 1 MHz". NEVER SELECT ANYTHING GREATER THAN 8 MHz
4. PROCESSOR: Select "ATTiny85"
5. PROGRAMMER: Select "USBTinyISP"

Load the program and select "Verify/ Compile" in the **SKETCH** tab. Then, with the programmer connected, select "Upload Using Programmer in the same tab. No errors should be apparent. LED flashes on programmer during programming. Takes about 3 seconds to program.

Background here:

[Tiny AVR Programmer Hookup Guide - SparkFun Learn](#)

[Download | USBtinyISP | Adafruit Learning System](#)